

Mechanistic Modeling of the KCNQ2 R581Q Mutation in Mouse and Human Layer 5 Pyramidal Neurons

Abstract—The KCNQ2 R581Q loss-of-function mutation reduces Kv7.2-mediated M-current and is associated with abnormal burst firing and altered action-potential and afterhyperpolarization properties. This project will build a biophysically detailed framework to test how R581Q changes intrinsic excitability in mouse and human layer 5 pyramidal neurons. A Hodgkin–Huxley Kv7.2 model will be fitted to experimental voltage-clamp data from Simkin et al., incorporated into validated mouse and human e-models on the Open Brain Institute platform, and evaluated through current-clamp simulations and electrophysiological feature extraction. A factorial Kv7.2/SK/BK conductance sweep will test whether Kv7.2 loss alone reproduces the mutant phenotype or whether compensatory SK/BK upregulation is also required. Cross-species analysis will determine whether the resulting excitability changes are conserved, amplified, or attenuated between mouse and human neurons.

Index Terms—KCNQ2, Kv7.2, R581Q, computational neuroscience, Hodgkin–Huxley model, neuronal excitability, channelopathy

I. INTRODUCTION AND RESEARCH QUESTION

Kv7.2 channels, encoded by *KCNQ2*, carry an important component of the neuronal M-current and help stabilize membrane potential, regulate action-potential firing, and limit repetitive activity. Loss-of-function *KCNQ2* mutations can cause developmental and epileptic encephalopathy. Simkin et al. showed that R581Q substantially reduces Kv7.2-mediated current and that affected human neurons exhibit abnormal burst firing together with altered action-potential repolarization and afterhyperpolarization properties [1].

A detailed single-neuron model provides a controlled way to separate the direct effect of Kv7.2 loss from other cellular changes that occur simultaneously in biological neurons. The central question is therefore: *How does the R581Q mutation in Kv7.2 alter the intrinsic excitability and firing properties of mouse and human layer 5 pyramidal neurons, and is this driven by Kv7.2 loss alone or by compensatory SK/BK upregulation?* Hodgkin–Huxley-style models are well suited to this problem because channel-level perturbations can be introduced explicitly and their effects on firing can be quantified [2].

II. DATA AND MODEL RESOURCES

Entire setup will be hosted on the OBI(Open Brain Institute) platform with additional processing on Google Colab pro Jupyter notebooks. The Kv7.2 ion channel will be modelled using Hodgkin-Huxley formalism. Additional fitting will be based on the dataset understanding. This channel will be added to validated mouse L5PC (from OBI platform) and human L5PC (from Blue Brain project) continuous adapting pyramidal (cADpyr) e-models. Simulation campaigns, parameter sweeps, and analysis will be run on the same setup. eFEL- Electrophysiology Feature Extraction Library for feature extraction also to be used extensively. SK and BK channels to be added in both cells using the same formalism in later phases.

Resulting values from Simkin et al.(2021) [1] to be used for building the Kv7.2 channels with the R581Q mutation and fitting the resulting L5PC models to the understanding derived from the firing pattern of the iPSC derived cortical neurons. The underlying principles of AP half-width, fAHP/mAHP/sAHP, RMP, and burst-firing metrics used for the ground-truth firing phenotype developed by Simkin et al will be adapted to a novel context of L5PC neurons. Furthermore, we will also consider compensatory channel (SK/BK channel) upregulation in R581Q neurons relative to WT. They will also be used for model fitting for the mutant phenotype.

III. METHODOLOGY

A. Kv7.2 Construction and Cell Integration

Kv7.2 will be represented using Hodgkin–Huxley(HH) formalism. The wild-type mechanism will first be fitted to the CHO voltage-clamp recordings, after which the R581Q condition will be represented using the mutation-associated current reduction and kinetic changes reported experimentally. Fitting will optimize parameters such as half-activation voltage $V_{1/2}$, slope factor k , and activation time constant τ across all available voltage steps. Fit quality will be evaluated using per-step current residuals, conductance-voltage agreement, activation-time-constant agreement, and biophysical sanity checks on the final parameters.

TABLE I
PLANNED CONDUCTANCE SWEEP.

Channel	Values
$g_{Kv7.2}$	100% (WT), 70%, 43% (R581Q), 20%
g_{SK}	100%, approx. 122% (week 3), 116–139% (week 5)
g_{BK}	100%, approx. 117% (week 3), 123% (week 5)

The fitted wild-type and mutant mechanisms will then be added to the mouse and human L5PC models. The same channel formulation and stimulation protocols will be used across species wherever the underlying model constraints permit. If SK or BK mechanisms are absent from a base model, they will be added before the compensation analysis.

B. Excitability and Model-to-Experiment Validation

Wild-type and mutant current-clamp simulation campaigns will be run and electrophysiological features will be extracted with eFEL. The planned features include rheobase, firing-rate versus current gain, input resistance, action-potential threshold, amplitude and half-width, fast/medium/slow afterhyperpolarization, spike-frequency adaptation, inter-spike-interval coefficient of variation, and burst index. For direct validation, simulated action-potential half-width, afterhyperpolarization measures, resting membrane potential, and input resistance will be benchmarked against Simkin et al. [1]. A single maturation timepoint will be used within each validation analysis rather than mixing developmental stages.

C. Factorial Sensitivity Analysis

A factorial conductance sweep will test whether Kv7.2 loss alone reproduces the mutant phenotype or whether SK/BK compensation is required. The main conditions are shown in Table I.

The 43% Kv7.2 condition follows the reported R581Q current-density reduction, while 70% and 20% provide exploratory bracketing points. SK/BK scaling follows the reported qPCR fold changes. Because different subunits change at different maturation stages, week-3 and week-5 compensation profiles will be simulated separately. Conductance scaling from qPCR is a modeling assumption and will not be interpreted as a directly measured conductance change.

D. Cross-Species Analysis

Identical stimulation protocols, temperature settings, and channel mechanisms will be applied to both species

wherever possible. For each feature f in species s , mutation-induced change will be calculated as

$$\Delta f_s(\%) = \frac{f_{\text{mutant},s} - f_{\text{WT},s}}{f_{\text{WT},s}} \times 100. \quad (1)$$

Mouse and human percentage changes will then be compared to determine whether the R581Q effect is conserved, amplified, or attenuated. A secondary analysis will test how conclusions change when homologous parameters are matched more tightly between the two species-specific models.

IV. EXPECTED CONTRIBUTIONS AND LIMITATIONS

The project has three main contributions: quantifying species-specific effects of Kv7.2 loss in mouse and human L5PCs; translating the Simkin et al. wet-lab findings into a reusable biophysical framework constrained at channel and neuronal levels; and testing causally whether Kv7.2 loss alone explains the abnormal firing phenotype or whether SK/BK compensation is required.

The main limitation is that SK/BK scaling comes from qPCR expression rather than direct conductance measurements. Cell-level validation is also human-only, so the mouse arm is a cross-species comparison rather than equivalent experimental validation. Sensitivity to differences between the underlying mouse and human base models will therefore be examined explicitly.

V. WORK PLAN AND CONCLUSION

Month 1 will cover OBI model inspection and data preparation; Months 2–3, Building Kv7.2 models (WT R581Q) with channel level validation, fit to Simkin et al data using OBI’s multi-objective evolutionary algorithm; Months 3–4, Model integration using HH-formulation for other ion channels, mouse/human L5PC integration ; Months 4–5, reproduction and validation of the mutant phenotype; and Months 5–6, the factorial sweep and cross-species analysis. The resulting framework will connect the R581Q channel defect to neuronal dysfunction while separating direct Kv7.2 loss from compensatory SK/BK effects.

REFERENCES

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